

Implementation of a Biomarker-Preserving Bone Biopsy Decalcification Workflow for Oncology Patients: A Single-Center Quality Improvement Initiative

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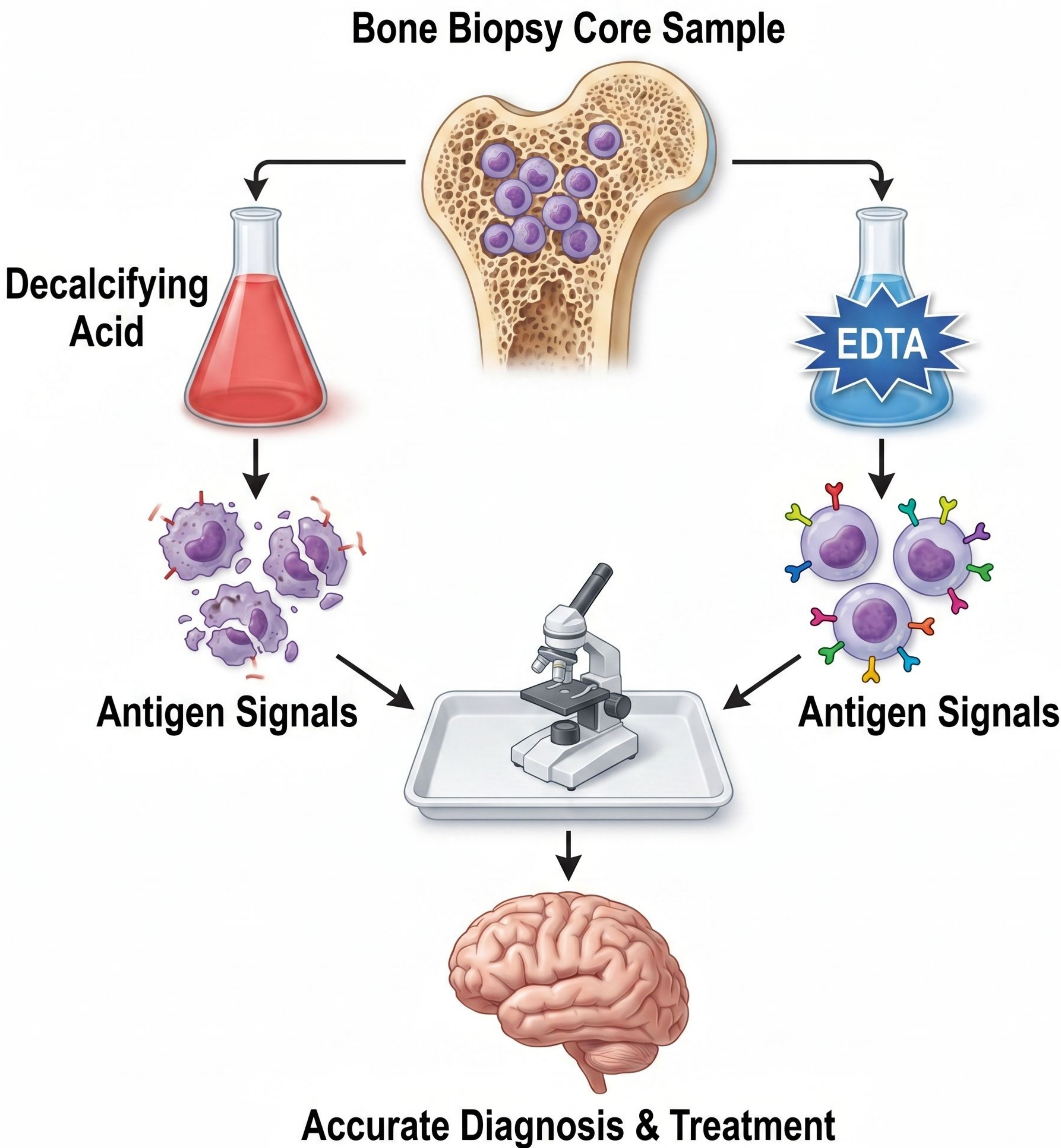
INTRODUCTION

Accurate histopathologic classification and biomarker assessment from bone metastasis biopsies are critical for guiding systemic therapy across solid tumors, including breast, prostate, lung, and urothelial cancers. Immunohistochemistry (IHC) and related biomarker analyses frequently determine treatment selection. However, osseous specimens typically require decalcification prior to evaluation, and preanalytic processing can significantly impact antigen preservation. Acid-based decalcification, commonly used in routine practice, may degrade antigenicity and increase the risk of false-negative or nondiagnostic biomarker results. In contrast, EDTA-based decalcification preserves receptor integrity but is not consistently implemented. Despite its clinical importance, there is often no standardized workflow to identify cases requiring biomarker-preserving processing. This gap represents a potentially modifiable source of diagnostic inaccuracy and suboptimal oncologic care.

OBJECTIVES

The primary objective is to design and implement a standardized, biomarker-preserving decalcification workflow for patients undergoing image-guided bone biopsy for suspected metastatic malignancy. Secondary objectives include:

- Evaluating baseline institutional decalcification practices and their impact on biomarker adequacy
- Identifying rates of nondiagnostic or discordant IHC results
- Assessing downstream clinical consequences, including repeat biopsies and treatment modifications



Anticipated Contributions

This study aims to establish a scalable, biomarker-preserving workflow for bone biopsy processing that improves diagnostic accuracy and enhances treatment selection. By addressing a key preanalytic variable, this initiative has the potential to reduce repeat procedures, improve patient outcomes, and standardize best practices across institutions.

Proposed Methods

This ongoing single-center QI initiative includes a retrospective baseline review of image-guided bone biopsies performed for suspected metastatic solid tumors between January 2023 and December 2025. Cases requiring treatment-informing IHC and/or lineage-defining biomarker evaluation are being assessed for decalcification method, adequacy of biomarker testing, discordance with known tumor biology or clinical presentation, need for repeat biopsy or outside pathology review, and downstream impact on treatment planning. Baseline institutional practice is being compared with published receptor-preserving standards. A multidisciplinary intervention was designed consisting of: (1) education of oncology, pathology, and interventional radiology; (2) a pathology workflow flag to identify specimens anticipated to require IHC; and (3) proposed EDTA-based decalcification for eligible cases.

Current Practice

Initial workflow review identified that current institutional practice applies routine acid-based decalcification to bone biopsy specimens without a standardized mechanism to differentiate cases anticipated to require treatment-informing immunohistochemistry (IHC). Based on current pathology workflow, 100% of decalcified bone biopsy specimens are processed using a standard acid-containing decalcification solution, while 0% are prospectively flagged for receptor- or biomarker-preserving processing when metastatic malignancy is suspected. This baseline process was recognized as a potentially modifiable preanalytic variable with implications for antigen preservation and downstream biomarker fidelity.